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① a) i. 4, ii. 4, iii. 4, iv. 3, v. 3, vi. 2, vii. 4.

b.) Reading A:- 4.7 mL sig figs (2)

Reading B:- 4.71 mL sig figs (3)

Ruler reading:- 2.20 cm sig figs (3)

② a) density = $\frac{\text{mass}}{\text{Volume}}$

$$\cancel{d = 58.26 - 65.41}$$

$$d = \frac{65.41 - 58.25}{5.00}$$

$$d = \frac{7.16}{5.00} = 1.432$$

$$d = 1.43 \text{ g/mL}$$

b) accepted density = 0.98 g/mL

$$\text{percent error} = \frac{1.43 - 0.98}{0.98} \times 100\%$$

$$= 45.918367.35$$

$$= 46\%$$

③ a) Gloves, Glasses (safety goggles).

b.) Go for eyewash station and inform your instructor.

- c. It should be thrown in the "Broken Glassware".

Flowchart :-

Measurements and Density Determination Part A:-

- ① At station A find the Volume measuring scales:-



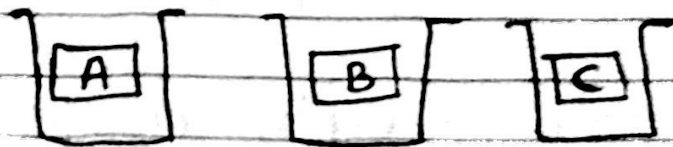
or



- ② Read the reading and record it.
- ③ Correct the significant figures.

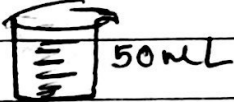
Part B:-

- ① At station B choose only one sample from A, B, C which are labeled.



- ② Place a ~~weighting~~ weighting boat on the top of the ~~loading~~ balance, then clean the pan before this step.
- ③ Press zero on the instrument to deduct the weight of the boat.

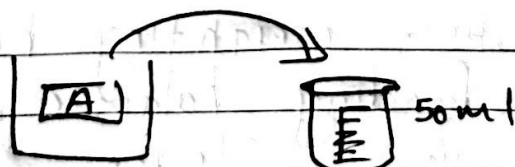
- ④ Add sample carefully to the weighing boat using spatula.
- ⑤ Make sure weight to between 1.50g to 2.00g of the sample.
- ⑥ Record the mass.
- ⑦ Leave the sample on boat for next step.
- ⑧ Press zero button on the instrument to make sure weight goes to 0.

- ⑨ Place 50mL Beaker on balance then close all doors. 

- ⑩ Record the mass of empty beaker.

- ⑪ Put the same beaker on benchtop.

- ⑫ Put the sample of part I into this Beaker.



- ⑬ Put beaker in the balance record data.



Part C:-

- ① At station C get equipment for the density.
- ② Record the mass of the clean dry 50 ml Erlenmeyer flask + stopper.
- ③ Put both on bench top.
- ④ Pipette 25.00 ml of sample, put it into flask the stopper it.
- ⑤ Reweight them with sample and record.
- ⑥ Record the temperature of sample

Part D:-

- ① At station D see the setup of diff substances.
- ② Record the observations for each.